RESEARCH LETTER

Does the choice of channel attention matter for few-label hyperspectral classification? A controlled comparison under random, standard and patch-disjoint splits

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ABSTRACT
Channel-attention blocks are often added to lightweight networks for hyperspectral image classification, and their gains are usually reported on random pixel splits, where overlapping patches leak spatial context into the test set. We ask whether block choice still matters under stricter evaluation. Seven modulation blocks (SE, ECA, CBAM, GCT, SRM and two variants of a dispersion-aware block we introduce, spectral-neighbourhood self-modulation, SNSM) are compared on one backbone with one protocol, no test-based model selection, up to fifteen paired seeds and Holm-corrected Wilcoxon tests. With random 5% splits, blocks differ only on Indian Pines, where SNSM raises average accuracy (AA) by 1.8–3.8 points over SE, ECA, CBAM, GCT and the plain CNN and ties with SRM, the other dispersion-based gate, through classes with one or two training pixels; overall accuracy differs by under 0.2 points. On the standard Indian Pines split all blocks lie within 0.35 points. On patch-disjoint splits of Pavia University and Kennedy Space Center (KSC), accuracy falls by 32–34 points and most block differences are not significant. We also show that the distributed KSC file stores small negative values as unsigned integers; reading it as signed raises seven convolutional variants from 77–92% to over 99% AA.

KEYWORDS
Hyperspectral image classification; channel attention; spatial leakage; evaluation protocol; few labels

1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Convolutional neural networks (CNNs) that operate on spatial–spectral patches are the usual lightweight choice (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019; Makantasis et al. 2015; Chen et al. 2016; Zhong et al. 2018; Roy et al. 2020), and transformers add global context at a higher cost (Hong et al. 2022; Sun et al. 2022). Channel attention is a cheap way to make such networks input-dependent: squeeze-and-excitation (SE) predicts one scale per channel from pooled means through fully connected (FC)

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layers (Hu, Shen, and Sun 2018); ECA replaces the FC layers by a one-dimensional convolution (Wang et al. 2020); CBAM adds a spatial gate (Woo et al. 2018); GCT normalises channel statistics and gates them with $1 + \tanh(\cdot)$ (Yang et al. 2020); and SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019). Hyperspectral networks have adopted these blocks widely (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021).

Most reported gains come from random pixel splits. With patch-based models, training and test patches then overlap, so test accuracy partly measures memorised spatial context (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). A recent survey documents this leakage (Feng et al. 2023), disjoint sampling changes the accuracy of deep models (Ahmad et al. 2024), and spatially separated splits lower the overall accuracy of CNNs and transformers on Pavia University by about 40 points (Alkhatib 2026). These studies compare architectures. Whether the *choice of channel-attention block* still matters once the protocol is tightened has, to our knowledge, not been tested with paired statistics.

Our contributions are:

- a controlled comparison of seven modulation blocks on one backbone, with identical inputs, schedule and splits, no test-based model selection, up to fifteen paired seeds and Holm-corrected Wilcoxon tests, under three protocols: random 5% splits, the standard fixed Indian Pines split, and patch-disjoint splits in which no test patch overlaps a training patch (Sections 2–3);
- the finding that block choice matters only in one regime, random splits of Indian Pines whose smallest classes have one or two training pixels, and that the advantage of dispersion-based gates there disappears under the other two protocols (Section 4);
- SNSM, a dispersion-aware block that matches the best baseline, with an ablation showing that the channel variance carries its gain;
- a correction of the Kennedy Space Center (KSC) data encoding, which earlier work treated as sensor saturation (Kakde et al. 2018), with its effect on CNN accuracy.

2. Blocks compared

All blocks are inserted in the same backbone. A 3×3 convolution maps the B input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each computing

$$\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{x}))), \qquad \mathbf{y} = \mathrm{ReLU}\big(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x}\big), \quad (1)$$

where $\mathcal{M}$ is the block under study ($\mathcal{M}$ = identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores. SE, ECA (kernel 3, the adaptive value for 64 channels) and CBAM (reduction 16, 7×7 spatial gate) output a sigmoid scale per channel; GCT uses its ℓ_2 embedding and an identity start; SRM gates each channel from its mean and standard deviation through a channel-wise FC layer and batch normalisation.

SNSM. For $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, SNSM computes for every channel c the spatial mean μ_c, the spatial variance v_c and the cross-channel gradient $g_c = (\mu_{c+1} - \mu_{c-1})/2$. Each descriptor vector is standardised across channels, the three are stacked into $\mathbf{S} \in \mathbb{R}^{3\times C}$,

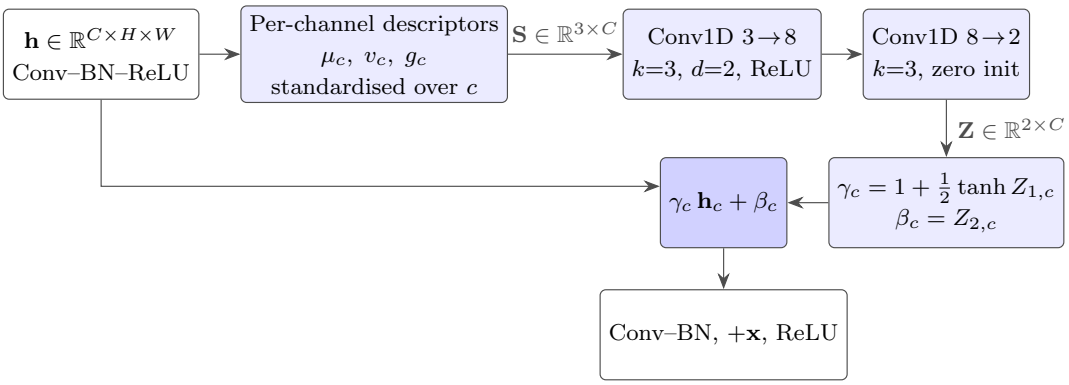


Figure 1. SNSM inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale γ_c and a shift β_c for each channel. The second convolution starts at zero, so the block is initially the identity.

and two one-dimensional convolutions along the channel axis (8 hidden channels, kernel 3, dilation 2 in the first, replicate padding) give $\mathbf{Z} \in \mathbb{R}^{2\times C}$. The modulation is

$$\gamma_c = 1 + \tfrac{1}{2}\tanh(Z_{1,c}), \qquad \beta_c = Z_{2,c}, \qquad \mathcal{M}(\mathbf{h})_c = \gamma_c\,\mathbf{h}_c + \beta_c. \tag{2}$$

The second convolution starts at zero, so the block is initially the identity; it adds 130 parameters per block (Figure 1). SNSM (γ) drops the shift. Like SRM it uses channel dispersion; unlike SRM it filters the descriptors jointly across neighbouring channels and can shift as well as scale. Because learned channels have no wavelength order, "neighbourhood" refers to adjacent feature channels.

3. Data and protocols

Data. We use Indian Pines (IP), Pavia University (PU), KSC and Botswana (BW) (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020) (Table 1). Each band is smoothed with a 5×5 Wiener filter, principal component analysis (PCA) (Jolliffe and Cadima 2016), fitted without labels on all pixels, keeps 30 components, and the db4 wavelet approximation of each component (Mallat 1989) is resized and appended, giving $B = 60$ features. Each labelled pixel is represented by its $15 \times 15 \times B$ patch.

KSC encoding. The commonly distributed KSC file is stored as unsigned 16-bit integers, but it contains small negative values (-27 to -1) that are stored in two's complement and are therefore read as 65,509–65,535; about 5% of its pixels contain at least one such value. Kakde et al. (2018) noticed these values, described them as saturation and set them to zero, but widely used toolboxes still load the file unchanged and scale it by its global minimum and maximum (Audebert, Le Saux, and Lefevre 2019), which compresses the valid range of 0–1,244 into less than 2% of the scale. We read KSC as signed integers. Without this correction, the wrapped values dominate the PCA and the seven CNN variants reach only 77–92% AA on KSC with random 5% splits, instead of 99.2–99.5%.

Protocols. *Random*: for each class, 5% of the labelled pixels (at least one) are drawn for training and the rest are used for testing, giving 505/9,744 (IP), 2,135/40,641 (PU), 256/4,955 (KSC) and 156/3,092 (BW) pixels; three IP classes then have one or two training pixels. *Standard*: the fixed IP training and test sets distributed with

Table 1. Benchmark scenes and the random 5% splits (seed 42; split sizes are the same for every seed).

Scene	Sensor	Size (pixels)	Bands	GSD (m)	Classes	Labelled	Train/test (random)
Indian Pines (IP)	AVIRIS	145×145	200	20	16	10,249	505/9,744
Pavia University (PU)	ROSIS	610×340	103	1.3	9	42,776	2,135/40,641
Kennedy Space Center (KSC)	AVIRIS	512×614	176	18	13	5,211	256/4,955
Botswana (BW)	Hyperion	1476×256	145	30	14	3,248	156/3,092



Figure 2. Patch-disjoint split of Pavia University (seed 42). Training pixels form one compact cluster per class; test pixels whose 15×15 patch would overlap a training patch are excluded.

SpectralFormer (Hong et al. 2022) (695/9,671 pixels; 15–50 training pixels per class). Its training pixels come from separate regions, but with 15×15 patches every test patch still overlaps a training patch. The corresponding Pavia University and Houston sets were no longer available from the authors' link. *Patch-disjoint* (PU, KSC): for each class, the 5% of its pixels nearest a random centre form a compact training cluster, and every test pixel within 14 pixels of any training pixel is discarded, so no test patch overlaps a training patch (Figure 2); 33,847–36,453 (PU) and 3,077–3,513 (KSC) test pixels remain, covering all 9 PU classes and at least 11 of the 13 KSC classes. The same rule leaves no test pixels for five IP classes, so we use the standard split for IP.

Training and statistics. Every model sees the same features and splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate 10^{-3}, weight decay 10^{-4}), step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, without augmentation or early stopping; the final model is evaluated once, so no validation or test data are used for selection. We use 15 seeds (42–56) for all eight backbone variants on IP (random and standard) and on the patch-disjoint splits, and five seeds elsewhere. We compare SNSM (γ+β) with each other block by paired same-seed differences and two-sided Wilcoxon signed-rank tests, Holm-corrected over the seven comparisons for each scene and protocol. Runs used PyTorch

Table 2. Random 5% splits: OA and AA (%, mean ± std). The eight backbone variants use 15 seeds on IP and five elsewhere; FC (γ+β), HybridSN (Roy et al. 2020) and SSFTT (Sun et al. 2022) use five. Best mean per column in bold.

Method	IP		KSC		PU		BW	
	OA	AA	OA	AA	OA	AA	OA	AA
Plain CNN	96.28±0.79	89.49±2.44	99.65±0.34	99.51±0.36	**99.59±0.07**	99.01±0.16	99.09±0.55	98.87±0.90
SE (γ)	96.34±0.62	90.04±2.18	99.61±0.28	99.43±0.38	99.53±0.19	98.80±0.56	98.95±0.82	98.76±1.07
ECA (γ)	96.28±0.68	90.87±2.53	99.61±0.47	99.40±0.66	99.58±0.10	99.00±0.20	**99.13±0.76**	**98.90±1.08**
CBAM	96.19±0.60	91.58±1.49	99.54±0.56	99.23±0.83	99.58±0.06	**99.11±0.18**	98.99±0.87	98.78±1.20
GCT	96.28±0.80	90.07±2.34	**99.69±0.26**	**99.55±0.28**	99.58±0.08	98.99±0.18	99.11±0.56	98.88±0.91
SRM	96.20±0.71	93.04±2.14	99.69±0.36	99.51±0.49	99.55±0.07	98.92±0.17	98.87±0.63	98.67±0.97
SNSM (γ)	96.17±0.73	92.89±2.00	99.69±0.26	99.50±0.39	99.49±0.05	98.91±0.16	99.10±0.62	98.89±0.96
SNSM (γ+β)	96.27±0.67	**93.33±1.68**	99.63±0.55	99.45±0.73	99.47±0.05	98.93±0.16	99.04±0.69	98.85±1.03
FC (γ+β)	96.24±0.52	92.51±1.60	99.63±0.40	99.48±0.47	99.51±0.06	98.88±0.14	98.80±0.70	98.63±0.98
HybridSN	88.30±4.91	73.64±8.28	95.00±3.02	91.20±5.32	97.84±1.52	95.46±4.12	77.16±37.92	75.56±38.31
SSFTT	**96.71±0.79**	92.35±1.81	99.38±0.28	99.15±0.32	–	–	–	–

Table 3. AA under the random protocol (R) and OA and AA under the held-out protocol (H): the standard split for IP and the patch-disjoint split for PU and KSC (%, mean ± std; 15 seeds except AA_R on PU and KSC, five seeds). The last row is the range of the eight block means.

Block	IP			PU			KSC		
	AA_R	OA_H	AA_H	AA_R	OA_H	AA_H	AA_R	OA_H	AA_H
Plain CNN	89.49±2.44	95.15±0.29	97.24±0.23	99.01±0.16	55.31±9.82	67.03±5.03	99.51±0.36	75.13±4.97	63.55±6.41
SE (γ)	90.04±2.18	94.88±0.46	97.25±0.26	98.80±0.56	55.91±9.73	**67.81±4.39**	99.43±0.38	76.79±4.44	**67.20±6.66**
ECA (γ)	90.87±2.53	94.92±0.33	97.15±0.22	99.00±0.20	56.30±9.17	67.73±4.43	99.40±0.66	76.40±4.81	65.15±5.86
CBAM	91.58±1.49	95.08±0.67	97.45±0.29	**99.11±0.18**	56.11±10.31	66.31±4.80	99.23±0.83	76.68±4.74	66.10±7.39
GCT	90.07±2.34	95.08±0.26	97.23±0.22	98.99±0.18	54.59±9.68	66.42±4.57	**99.55±0.28**	75.20±5.16	63.49±6.80
SRM	93.04±2.14	95.20±0.54	**97.50±0.22**	98.92±0.17	56.02±9.99	66.00±4.73	99.51±0.49	74.36±4.22	62.75±6.38
SNSM (γ)	92.89±2.00	95.07±0.52	97.37±0.22	98.91±0.16	55.41±9.44	67.31±4.57	99.50±0.39	76.86±4.19	65.60±7.15
SNSM (γ+β)	**93.33±1.68**	**95.23±0.42**	97.48±0.26	98.93±0.16	**56.80±11.83**	67.34±4.95	99.45±0.73	**77.11±3.54**	67.01±6.00
Spread of block means	3.84	0.34	0.35	0.31	2.21	1.80	0.32	2.75	4.44

(Paszke et al. 2019) on NVIDIA Tesla T4 GPUs, except ten KSC patch-disjoint seeds, which ran on an RTX 3050 laptop GPU.

4. Results

Random splits. On PU, KSC and BW the eight backbone variants lie within 0.32 AA points of each other (Table 2). Only IP separates them: SNSM (γ+β) has the highest AA, and it is significantly better than the plain CNN, SE, ECA, CBAM and GCT (by 1.75–3.84 points; Table 4, first column), but not than SRM or SNSM (γ). The overall accuracy of all eight variants differs by only 0.17 points. The gain therefore comes from a few classes: Oats and Grass-pasture-mowed have one training pixel and 19 and 27 test pixels, so a change of a few test pixels moves their per-class accuracy by tens of points. The two blocks that see channel dispersion, SRM and SNSM, are the two best and are statistically tied. The externally designed HybridSN and SSFTT, trained under our protocol, are less accurate than the backbone variants.

Standard and patch-disjoint splits. With the standard IP split, every class has at least 15 training pixels and all eight blocks reach 97.15–97.50% AA (Table 3); the only significant difference left, against ECA, is 0.33 points. The 3.84-point spread of the random protocol thus does not survive a split that gives the small classes enough training pixels. On the patch-disjoint splits accuracy collapses: AA falls by 32.0 points on PU and 34.3 on KSC, averaged over blocks, and OA falls to 55–57% and 74–77%. Block means now spread by 1.80 (PU) and 4.44 (KSC) points, but the seed-to-seed standard deviation is 4.4–7.4 points. On PU no difference is significant; on KSC, SNSM (γ+β) is significantly better than the plain CNN, GCT and SRM, but not than SE, ECA, CBAM or SNSM (γ), and SE has the highest mean AA (Figure 3). The ranking of blocks also changes between protocols: SRM, tied best on random IP, is the lowest on patch-disjoint KSC.

Table 4. Paired AA differences SNSM (γ+β) − block (points; seeds won in parentheses). *: Holm-corrected two-sided Wilcoxon $p < 0.05$.

SNSM (γ+β) −	IP, random	IP, standard	PU, disjoint	KSC, disjoint
Plain CNN	+3.84* (14/15)	+0.24 (11/15)	+0.31 (8/15)	+3.46* (11/15)
SE (γ)	+3.29* (15/15)	+0.23 (11/15)	-0.47 (7/15)	-0.19 (8/15)
ECA (γ)	+2.45* (11/15)	+0.33* (13/15)	-0.39 (5/15)	+1.86 (11/15)
CBAM	+1.75* (13/15)	+0.04 (7/15)	+1.03 (10/15)	+0.91 (7/15)
GCT	+3.26* (14/15)	+0.25 (11/15)	+0.92 (9/15)	+3.51* (12/15)
SRM	+0.29 (8/15)	-0.02 (8/15)	+1.33 (11/15)	+4.25* (12/15)
SNSM (γ)	+0.44 (9/15)	+0.12 (10/15)	+0.03 (8/15)	+1.40 (9/15)

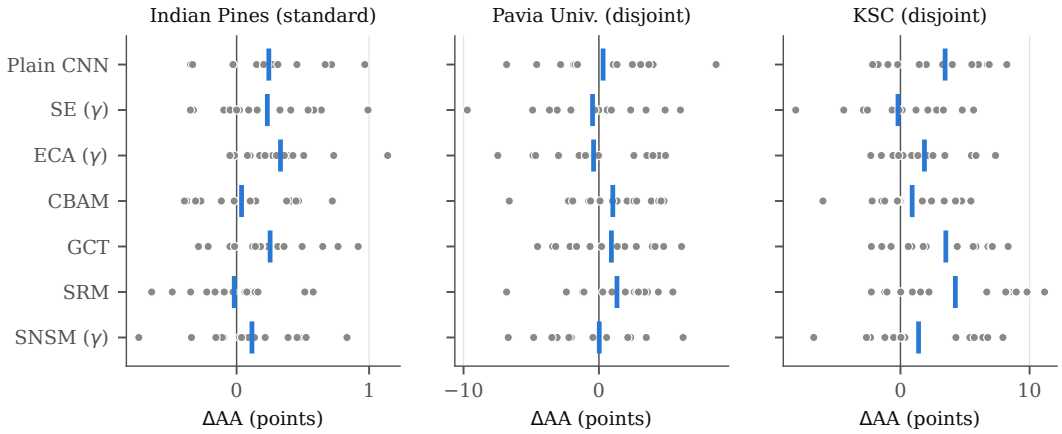


Figure 3. Paired same-seed AA differences SNSM (γ+β) − block under the held-out protocols. Grey dots are the 15 seeds; the blue bar is their mean.

What drives the gain on random IP. Removing descriptors from SNSM on random IP (five seeds; Table 5) shows that the variance carries the gain: with the mean alone AA falls by 1.16 points, and adding the gradient to the mean does not recover it; without the zero initialisation AA is lower on all five seeds. This matches the tie between SNSM and SRM, which also gates on dispersion. When SNSM is applied to the raw bands in wavelength order instead of to feature channels, it also raises IP AA by 3.9–4.8 points over SE, ECA and GCT applied the same way (five seeds, five wins each), but this raw-band pipeline is weaker overall and ties on PU and KSC (results in the repository).

Cost. The channel blocks add between 6 (ECA) and 4,256 (SE) parameters to the plain CNN, and SNSM adds 260 (Table 6). SNSM's descriptor computation consists of many small operations, so its batch-1 latency is about 3.7 times and its training time about 1.7 times that of the plain CNN.

5. Discussion and limitations

The choice of channel-attention block changed AA by more than half a point only when the smallest classes had one or two training pixels and test patches overlapped training patches. In that regime, the two gates that see channel dispersion (SRM, SNSM) were clearly better than SE, ECA, CBAM and GCT. Once the small classes had enough training pixels, or once no test patch overlapped a training patch, the blocks were practically or statistically indistinguishable, while the protocol itself moved AA by

Table 5. SNSM (γ+β) ablation on IP, random 5% splits (AA, %; five seeds). ΔAA is the paired difference from the full block; wins count seeds on which the variant is better.

Descriptors	AA	ΔAA (wins/5)
mean, var, grad (SNSM)	94.16±1.07	–
mean	93.01±0.72	-1.16 (1/5)
mean, var	94.55±1.67	+0.39 (4/5)
mean, grad	93.37±1.09	-0.80 (1/5)
mean, var, grad; default init.	92.67±1.15	-1.49 (0/5)

Table 6. Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs, random splits).

Method	Params (M)	MMACs[a]	Latency (ms)[b]	Train IP (s)	Train PU (s)
Plain CNN	0.184	41.2	0.69	4.2	17.6
SE (γ)	0.188	41.2	1.17	4.5	19.3
ECA (γ)	0.184	41.2	0.98	4.8	19.8
CBAM	0.185	41.3	1.55	5.9	24.9
GCT	0.184	41.2	1.11	4.7	20.0
SRM	0.184	41.2	1.22	4.8	20.2
SNSM (γ)	0.184	41.3	2.54	7.1	29.5
SNSM (γ+β)	0.184	41.3	2.54	7.2	30.2
FC (γ+β)	0.190	41.2	1.31	4.7	19.8
HybridSN	1.743	137.4	0.71	26.2	115.3
SSFTT	0.291	34.9	1.25	7.6	–

[a] Multiply–accumulates of convolution and linear layers counted by `thop`; tensor operations such as attention products and descriptor statistics are not included. [b] Batch size 1 on a Tesla T4.

more than 30 points. Gains of a few AA points reported for a new block on a random split should therefore not be read as evidence of better generalisation.

Several limitations apply. The standard IP split still lets test patches overlap training patches, and we could not obtain the standard PU and Houston splits; our patch-disjoint construction is one of several possible and leaves few test pixels for some KSC classes. PCA is fitted on all pixels of each scene without labels. All blocks share one backbone, input pipeline and schedule, none was tuned to it, and HybridSN and SSFTT are less accurate here than in their original papers. Wilcoxon tests with 15 seeds detect consistent shifts but not their practical relevance, so we report effect sizes alongside them.

6. Conclusion

Under one strict protocol, the choice among seven lightweight channel-attention blocks matters little for few-label hyperspectral classification. Dispersion-based gates, including the proposed SNSM, add 2–4 AA points on random Indian Pines splits through classes with one or two training pixels, but this advantage vanishes on the standard split and is mostly not significant on patch-disjoint splits, where accuracy drops by 32–34 points for every block. Evaluation protocol and data handling, such as the KSC encoding corrected here, affect reported accuracy far more than the attention block.

Disclosure statement

The authors report there are no competing interests to declare.

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Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020); the standard IP split is distributed with SpectralFormer (Hong et al. 2022). Code, per-run results and the scripts that generate every table and figure are available at `https://github.com/Devguru-codes/research_paper_code`.

Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

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